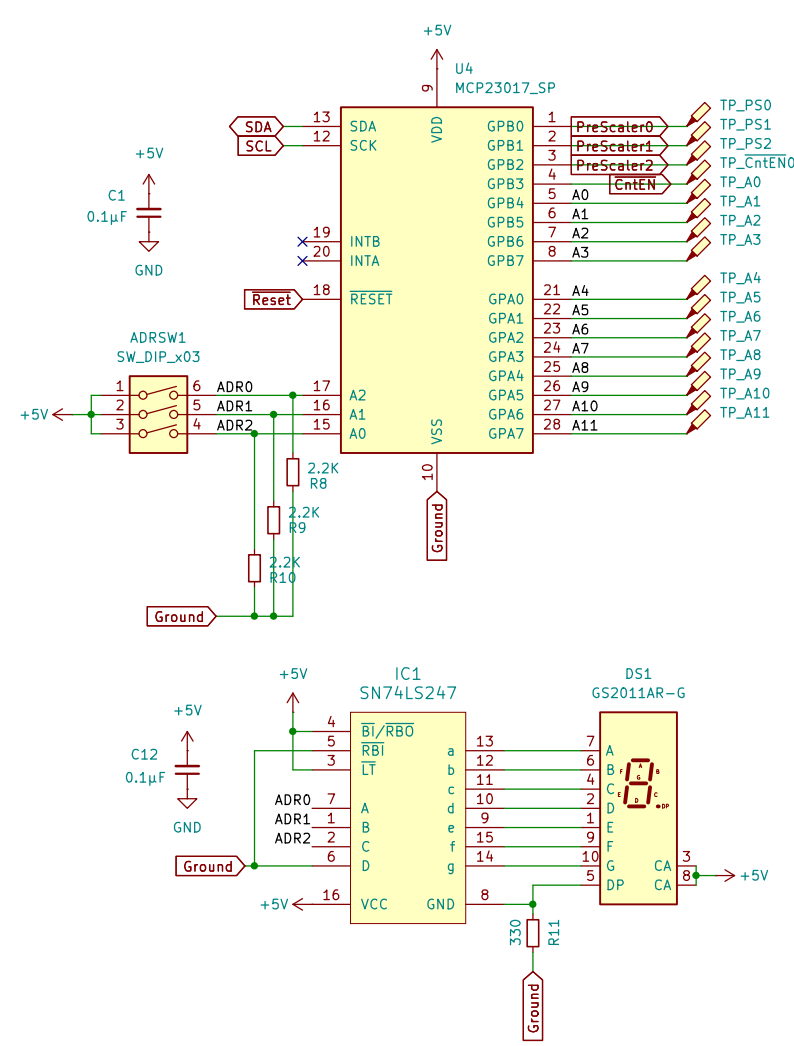
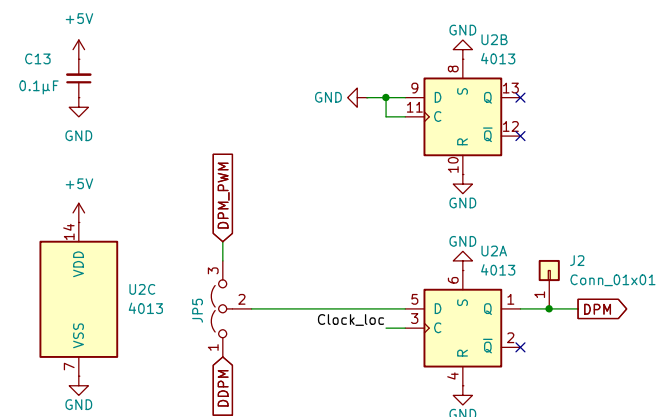
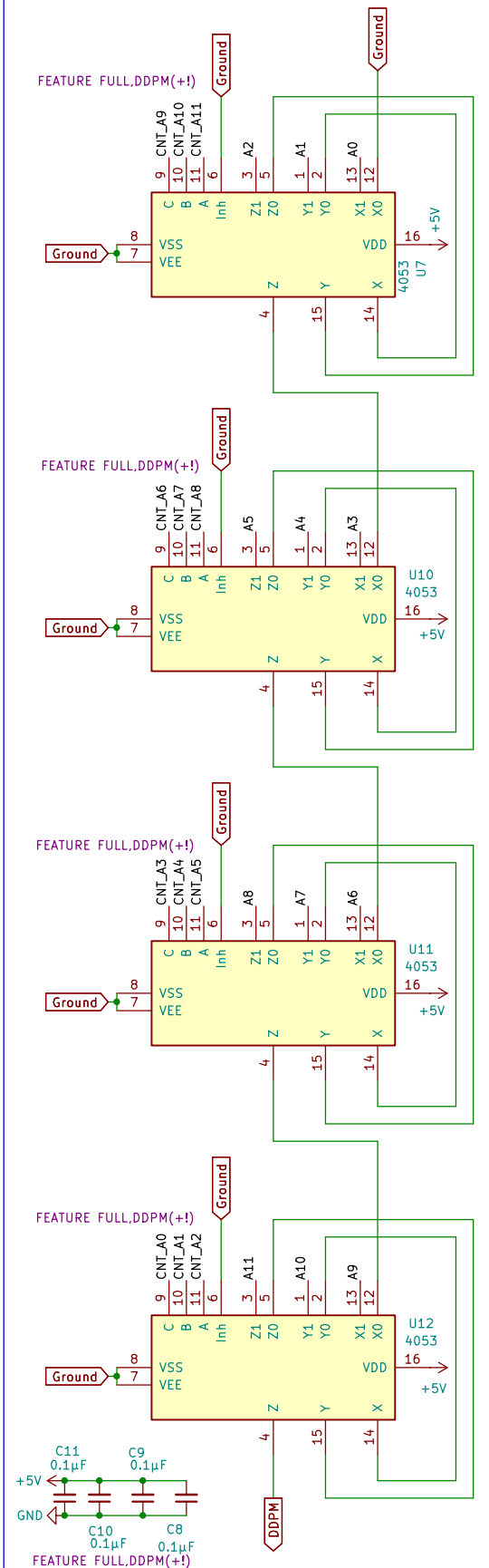
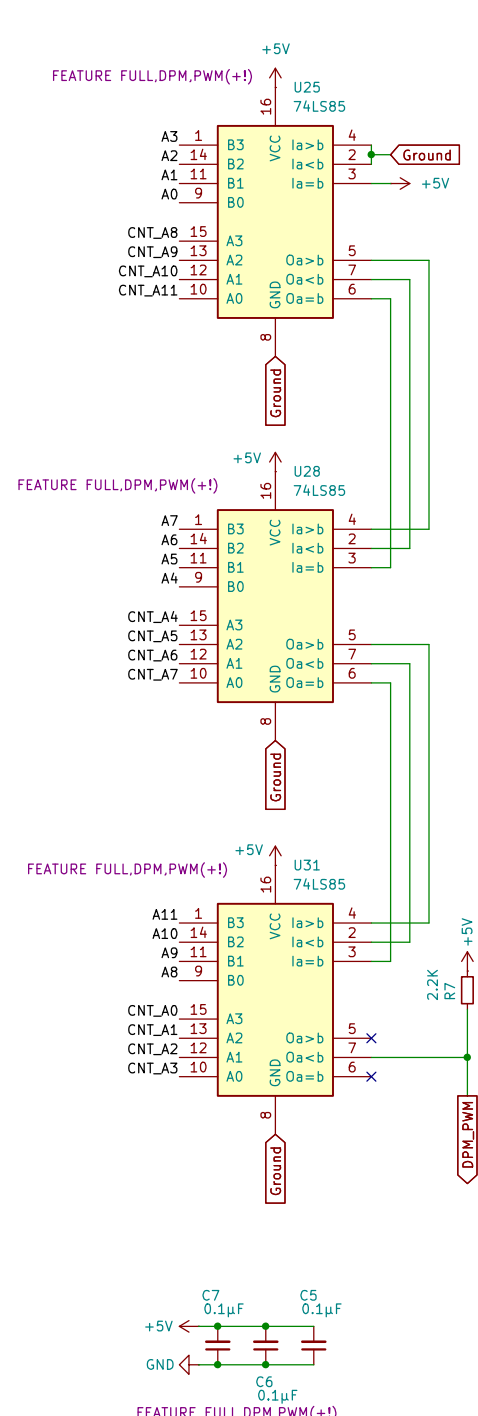
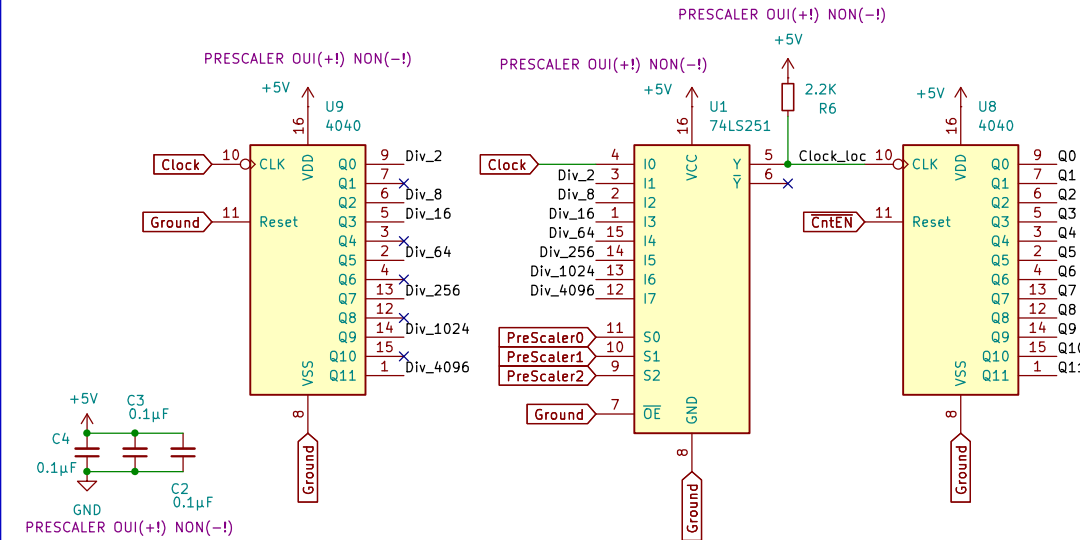
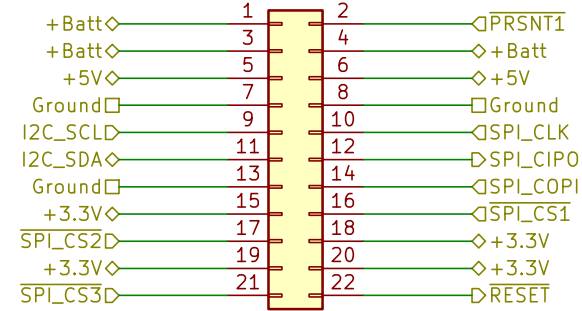
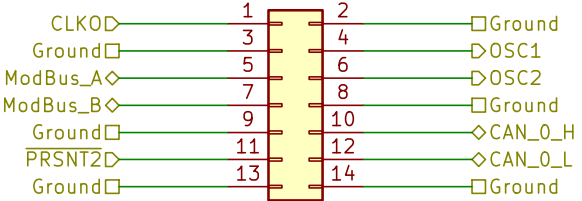


PCIe_SPI_1
Conn_02x11_Odd_Even



PCIe_CAN_1
Conn_02x07_Odd_Even



PCI Express connector pinout (x1, 5)			
Pin	Side B	Side A	Description
1	Vbatt	PRSNT1#	Must connect to farthest PRSNT2# pin
2	Vbatt	Vbatt	
3	+5v	+5v	Main power pins
4	GND	GND	
5	I2C_SCL	SPI_CLK	
6	I2C_SDA	SPI_CIPO	
7	GND	SPI_COPI	
8	+3.3v	SPI_CS1#	
9	SPI_CS2#	+3.3v	
10	+3.3v	+3.3v	
11	SPI_CS3#	RESET#	
12	CLK0	GND	
13	GND	OSC1	
14	CAN_1_H	OSC2	
15	CAN_1_L	GND	
16	GND	CAN_0_H	
17	PRSNT2#	CAN_0_L	
18	GND	GND	

PCI Express connector pinout (x1, 5)			
Pin	Side B	Side A	Description
1	+12 V	PRSNT1#	Must connect to farthest PRSNT2# pin
2	+12 V	+12 V	
3	+12 V	+12 V	Main power pins
4	Ground	Ground	
5	SMCLK	TCK	
6	SMDAT	TDI	
7	Ground	TDO	SMBus and JTAG port pins
8	+3.3 V	TMS	
9	TRST#	+3.3 V	
10	+3.3 V aux	+3.3 V	Standby power
11	WAKE#	PERST#	Link reactivation; fundamental reset [24]
Key notch			
12	CLKREQ# [24]	Ground	Clock Request Signal
13	Ground	REFCLK-	Reference clock differential pair
14	HSOp(0)	REFCLK+	Lane 0 transmit data, + and -
15	HSOn(0)	Ground	
16	Ground	HSIp(0)	
17	PRSNT2#	HSIn(0)	Lane 0 receive data, + and -
18	Ground	Ground	

PCI Express x1 cards end at pin 18

Liaison :
- PRSNT1# et PRSNT2# par le GNDD (équivalent présence carte PCIx1)

Remplacement :
- SMBus -> I2C
- JTAG -> SPI dont SC1# et SC2#
- WAKE# -> CS3#
- CLKREQ -> CLK0
- RefClk différentielle PCI -> OSC1 et OSC2 pour external clock MCU
- HSOp(0) soit TX0 PCI -> ModBus_A
- HSOn(0) soit TX0 PCI -> ModBus_B
- HSIp(0) soit RX0 PCI -> CAN_0_H
- HSIn(0) soit RX0 PCI -> CAN_0_L

PS :
L'utilisation d'un RS485 4 fils est possibles en cablant :
- HSOp(0) soit TX0 PCI -> DI_A master / RO_A slave
- HSOn(0) soit TX0 PCI -> DI_B master / RO_B slave
- HSIp(0) soit RX0 PCI -> RO_A master / DI_A slave
- HSIn(0) soit RX0 PCI -> RO_B master / DI_B slave

L'utilisation d'un RS485 2 fils est possibles en cablant :
- HSOp(0) soit TX0 PCI -> A master / A slave
- HSOn(0) soit TX0 PCI -> B master / B slave
- HSIp(0) soit RX0 PCI -> NC. ou CAN_0_H
- HSIn(0) soit RX0 PCI -> NC. ou CAN_0_L